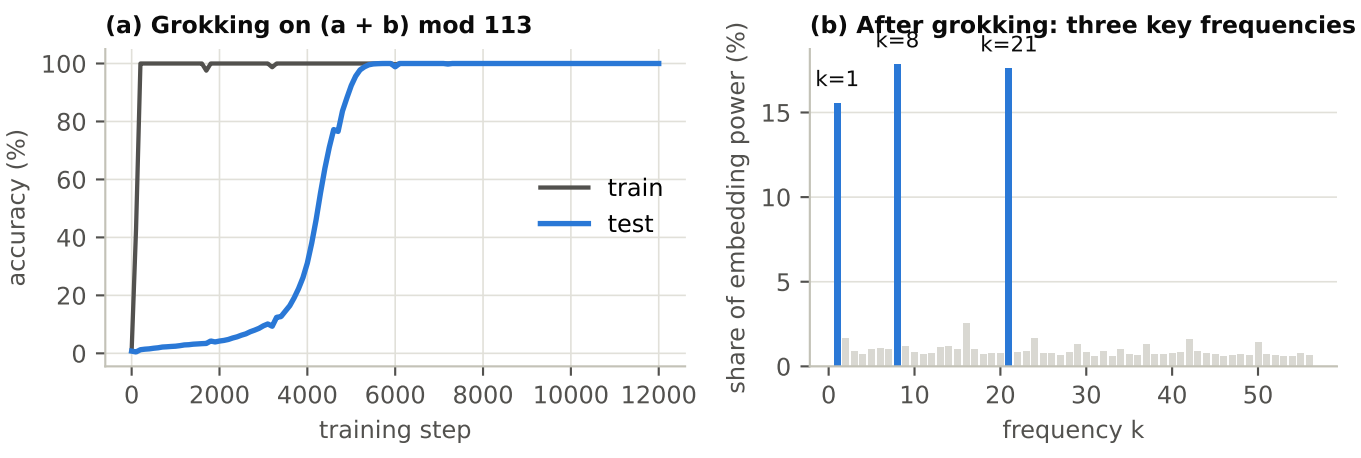
(a) Grokking on (a + b) mod 113
accuracy (%)
100
80
60
40
20
0
0
2000
4000
6000
8000
10000
12000
training step
train
test
(b) After grokking: three key frequencies
share of embedding power (%)
15
10
5
0
k=1
k=8
k=21
0
10
20
30
40
50
frequency k